

TITLE PAGE

Problem Statement ID – SIH26178

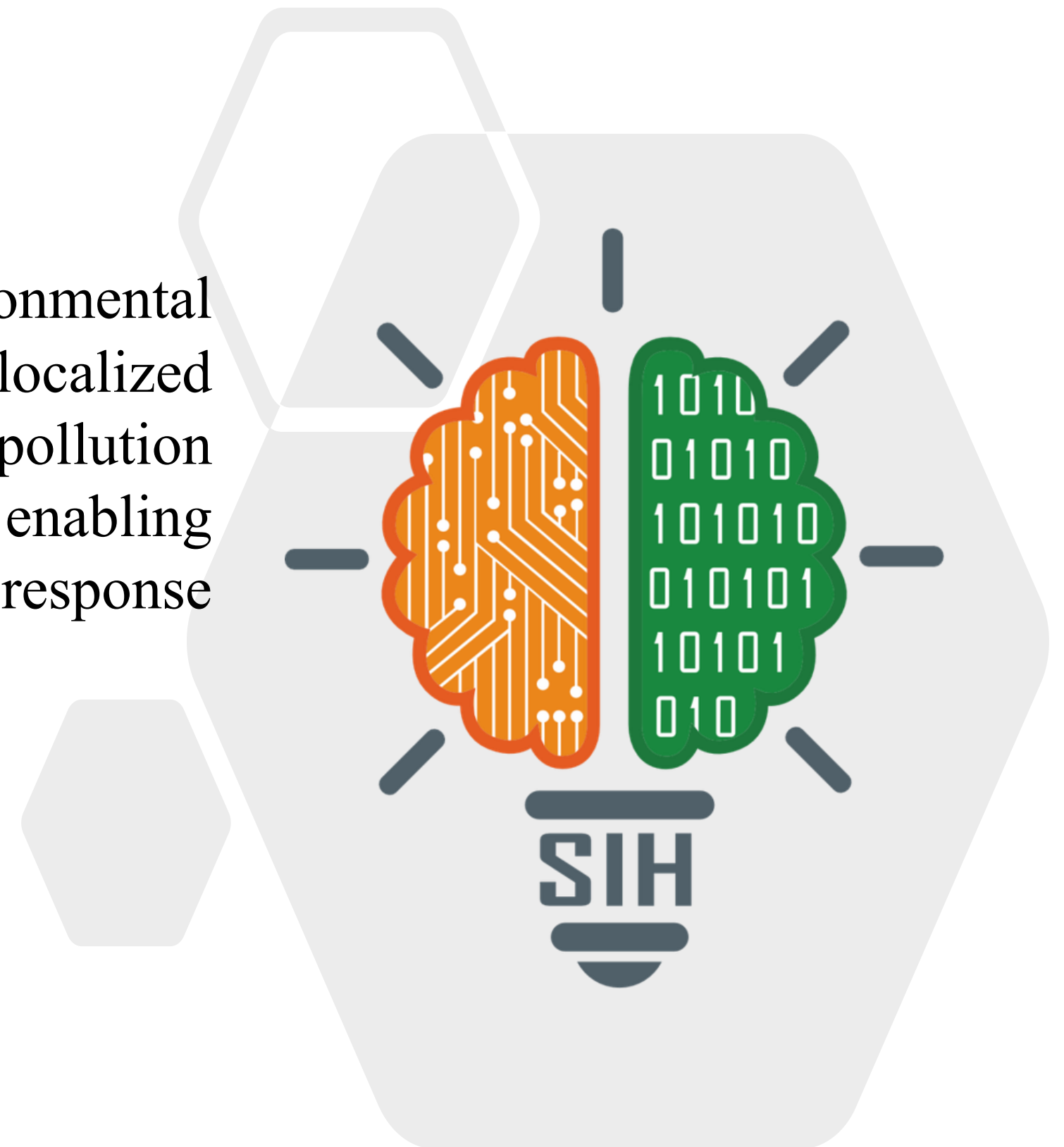
Problem Statement Title- A resilient, AI-powered environmental monitoring network that provides early detection, localized intelligence, and actionable alerts for floods, forest fires, pollution events, and other environmental hazards common in India, enabling authorities and communities to shift from reactive disaster response to proactive risk prevention.

Theme-Disaster Management

PS Category- Hardware

Team ID- 154320

Team Name- GEN ERA



THE CHALLENGE

Uttarakhand's extensive and geographically dispersed forest landscape makes it difficult to continuously monitor. A localized event can begin in remote areas and quickly escalate, leading to:



Forest Fires

Rapid spread, loss of forest cover and biodiversity



Air Pollution (PM2.5)

Degraded air quality and health risks



Gas Leakages

Risk to communities, infrastructure and responders

Our Approach: Deploy a localized, solar-powered IoT sensor network with Edge AI to continuously monitor multiple hazards, verify threats at the source, precisely map incidents, and trigger early alerts—even in remote, low-connectivity regions.

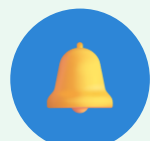
PROPOSED SOLUTION

A localized, AI-powered early-warning and threat mapping system



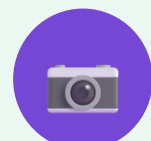
Sense Continuously

Monitor environmental conditions in real-time



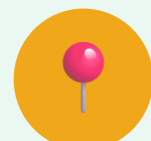
Identify Abnormal Conditions

Detect potential hazards early



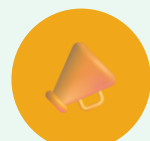
Verify the Threat

Visual confirmation, to reduce false alarms



Locate it

Geotag the incident for precise mapping



Alert Earlier

Enable faster and informed response

Detect closer to the source. Verify before escalation. Map the threat. Alert earlier.



Vast Forest Ecosystems

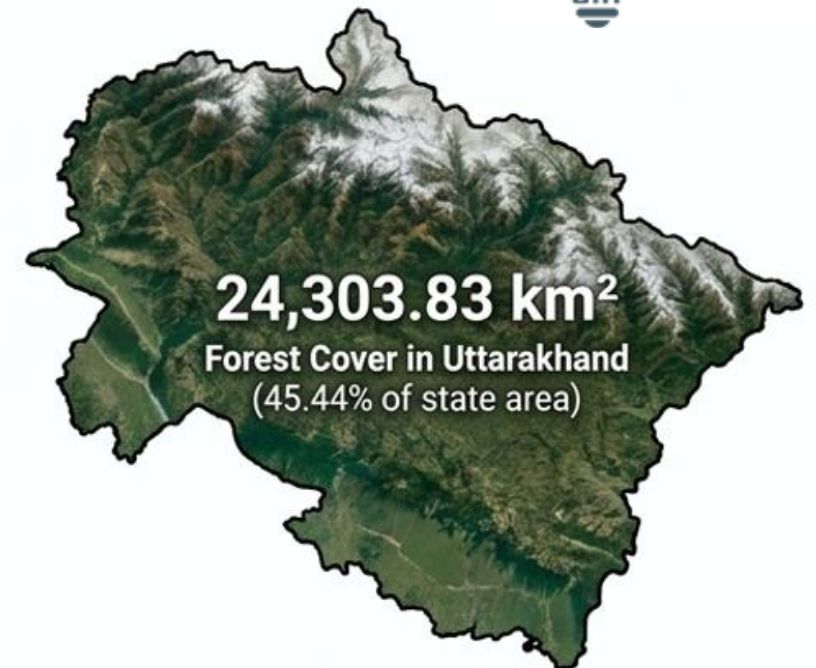


Remote & Dispersed Regions

Challenging Terrain



Prone to Multiple Environmental Hazards



OUR NOVELTY



Dynamic Power Management

Solar-powered with intelligence power mitigation to ensure long-term, off-grid operation.



Offline-First Operation

Works in remote areas with no internet; local edge AI processing and LoRa-based communication.



Model Drift Mitigation

Continuous sensor-based monitoring, periodic calibration and baseline comparison to reduce AI model drift.

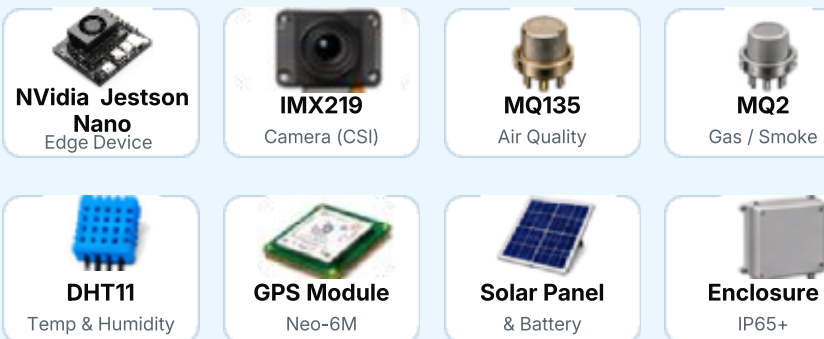


Geo-Mapping & Threat Intelligence

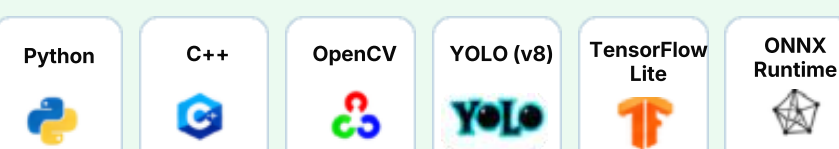
GPS-enabled incident mapping with precise location, severity and real-time dashboard visualization.

1 TECHNOLOGIES & TECH STACK

Edge Hardware & Sensors



Software & AI Frameworks



Communication & IoT Protocols



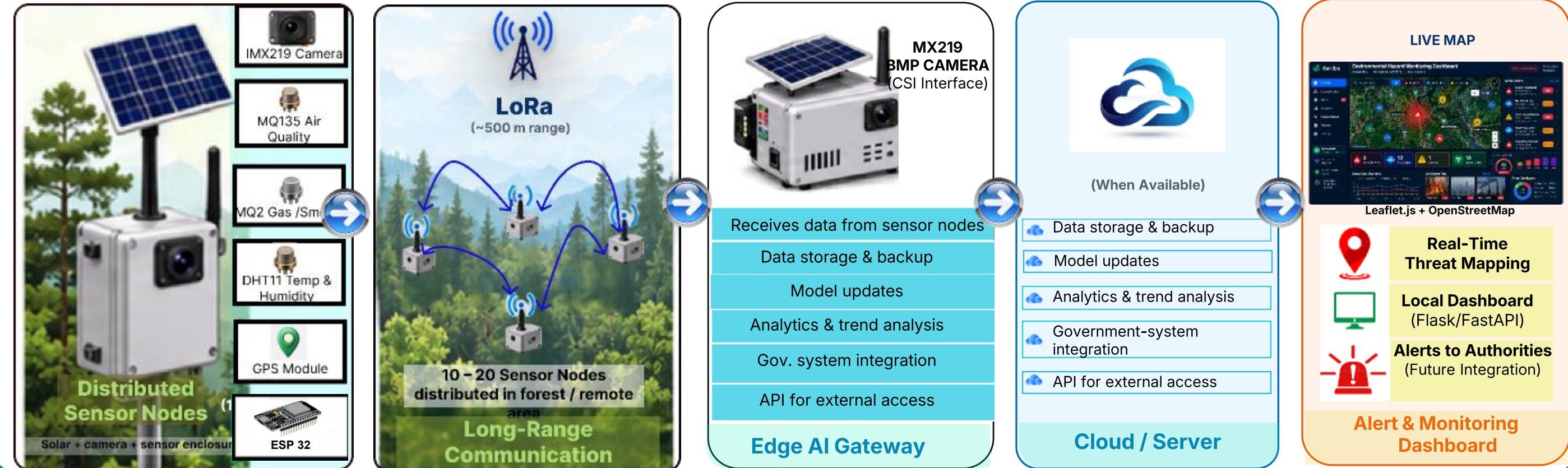
Backend & Web GIS



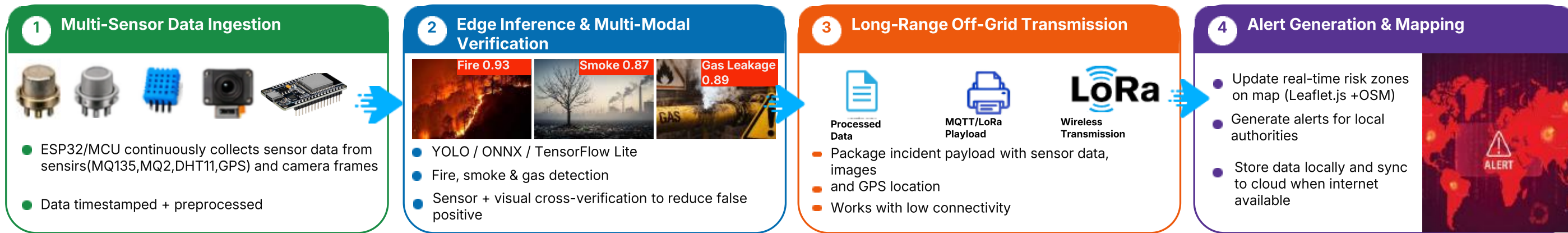
Power Management

Solar powered with adaptive power management for remote locations.

2 SYSTEM ARCHITECTURE



3 METHODOLOGY & IMPLEMENTATION WORKFLOW



4 AI MODELS & ALGORITHMS



5 DEPLOYMENT STRATEGY

- Offline-first edge devices
- Solar powered + low-maintenance
- Scalable multi-node deployment

6 KEY OUTCOMES

- Early detection of forest fire, air pollution and gas leakage
- Accurate location with GPS tagging
- Real-time alerts to authorities
- Cost-effective, scalable deployment

1. ECONOMIC FEASIBILITY

Sensor Node (ESP32-S3 with Camera)



Component (per sensor node)	Est. Cost (₹)
ESP32-S3 (Wi-Fi + BLE)	800
Environmental Sensors (MQ135, MQ2, DHT11, GP2Y1010)	1,000
GPS Module	600
LoRa Module (SX1278)	600
Camera (IMX219)	800
Solar Panel + Battery + Charge Ctrl.	800
Enclosure, Antenna & Misc	400
Total (per sensor node)	~ 5,000

- ✓ Each sensor node has its own camera for local visual capture and AI verification.
- ✓ Powered by solar + battery for off-grid operation.

Jetson Gateway (No Camera)

Component (gateway)	Est. Cost (₹)
NVIDIA Jetson Nano B01 (4GB)	8,000
LoRa Module + Antenna	1,500
Power System (Solar + Battery)	3,000
Weatherproof Enclosure + Misc	2,500
Total (per gateway)	~ 15,000

- ✓ Can connect to multiple sensor nodes (10 – 20 per gateway, depending on range)
- ✓ Processes data and runs AI inference.

Prototype Cost

(1 Gateway + 1 Sensor Node)

Total Prototype Cost
~ ₹20,000

- ✓ 1 Jetson Nano Gateway
- ✓ Complete, working prototype for field testing
- ✓ 1 ESP32-S3 Sensor Node (with camera + sensors)

2. DEPLOYMENT EXAMPLE: NAINITAL (UTTARAKHAND)

Nainital District (FSI 2023)

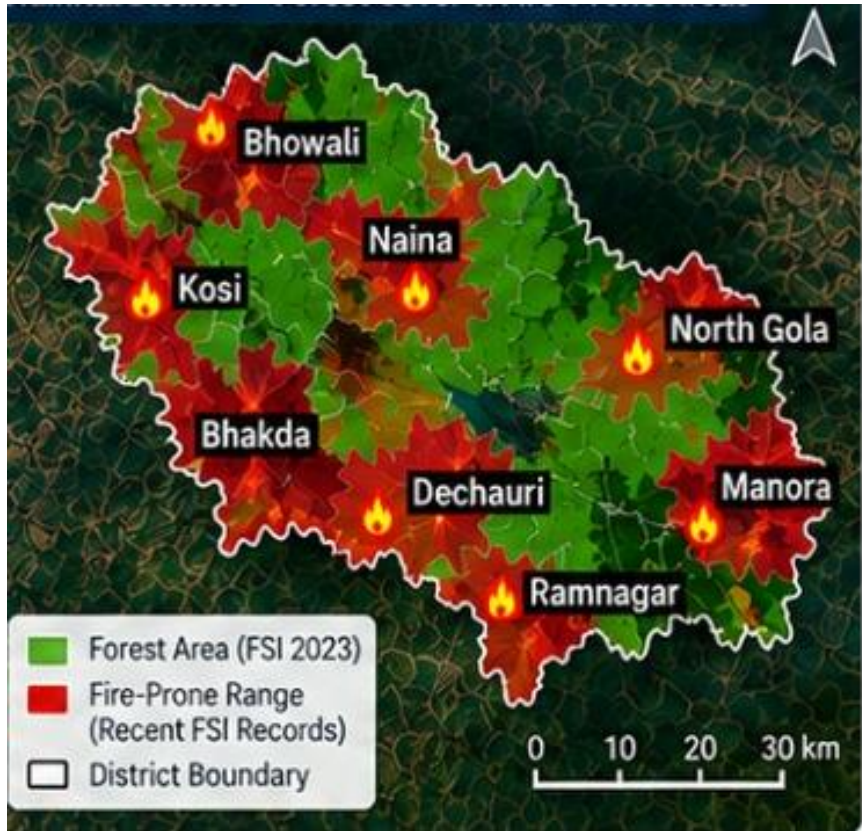
Forest Cover
2,866.55 km²
(67.43% of district area)

District Area
4,252.85 km²

Major Fire-Prone Ranges
(Recent FSI Fire Records)

- Bhowali Range
- Naina Range
- Kosi Range
- North Gola Range
- Dechauri Range
- Bhakda Range
- Ramnagar Range
- Manora Range

Nainital District – Forest Cover & Fire-Prone Areas



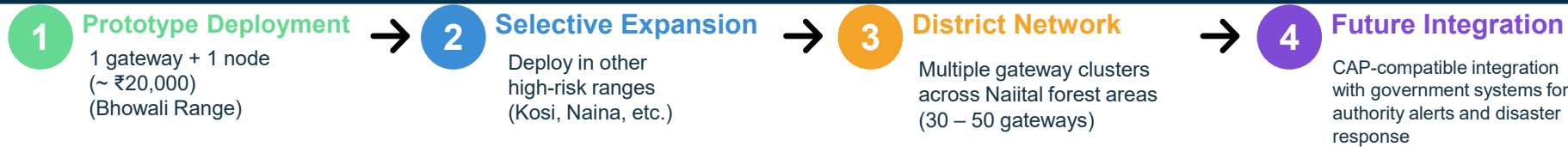
3. KEY RISKS & CHALLENGES

Risk / Challenge	Impact	Mitigation Strategy
Jetson Compute Load	Multiple sensor streams can increase processing demand.	Event-Triggered AI • Activate vision inference only when risk detected • Optimized model (TFLite / ONNX)
LoRa Packet Loss	Terrain, interference and distance can cause missing packets from sensor nodes.	Reliable LoRa Communication • Acknowledgement & retry mechanism • Timestamped data buffering
Sensor Calibration Drift	Sensor readings may change over time due to ageing, environmental conditions and dust/moisture.	Sensor Health & Calibration • Periodic calibration and compensation • Multi-model verification (sensor + vision)

4. SCALABILITY ANALYSIS – FROM PROTOTYPE TO DISTRICT-WIDE

Deployment Level	Coverage / Area (approx.)	Gateways (Jetson)	Sensor Nodes (per gateway)	Total Nodes	Estimated Cost (₹)
Prototype (Bhowali Range)	~ 5 – 10 km ² (high-risk zone)	1	1	1	~ 20,000
Small Cluster	~ 50 – 100 km ²	1	10 – 20	10 – 20	~ 65,000 – 1.15 lakh
Priority Zones (8 major ranges)	~ 50 – 100 km ² each (combined)	8	10 – 20	80 – 160	~ 5.2 – 9.2 lakh
Selective District Deployment	Selected forest areas (~ 10 – 30% of forest cover)	30 – 50	10 – 20	300 – 1,000	~ 19.5 – 57.5 lakh

5. DEPLOYMENT PATH & LONG-TERM VIABILITY



1. POTENTIAL IMPACT ON TARGET AUDIENCE



Disaster Response Teams (NDRF / First Responders)

- ⚡ Accelerates emergency dispatch by providing real-time hazard coordinates and high-precision spatial risk boundaries.
(target: 1 – 2 sec local alert latency).
- Helps in ground coordination with live location, affected area and risk intensity.



District Administration & Local Authorities

- 📊 Automates district-wide threat tracking with real-time dashboard and analytics.
- 📋 Supports regulatory and compliance reporting.
- ➦ Designed to be CAP-compatible for future integration with SANCHET and other government alert systems.



Local Populations & Forest Border Communities

- 🚩 Delivers early warnings for approaching forest fires, gas leakages and hazardous air quality spikes.
- ♣️ Enables timely evacuations and better preparedness, safeguarding human life in vulnerable areas.
- 🏠 Builds community resilience through real-time, easy-to-understand alerts (mobile/dashboard).



2. MULTI-DIMENSIONAL BENEFITS OF THE SOLUTION



SOCIAL BENEFITS



Enables early warnings and evacuation support during forest fires, gas leakages and toxic PM2.5 events, reducing risk to human health and life.



Helps reduce exposure

to hazardous air pollution, lowering risk of respiratory illnesses.



Enhances public safety and community resilience in low-connectivity rural and forest regions.



ECONOMIC BENEFITS



Minimizes financial losses for industries, infrastructure and forest resources by detecting gas leaks and localized fires before escalation.

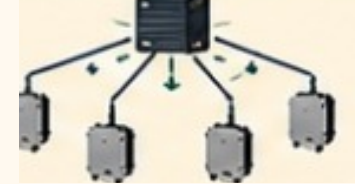


Cost-effective deployment:

Prototype (1 Jetson Gateway + 1 Sensor Node) ≈ ₹20,000.



One gateway can serve multiple sensor nodes (10 – 20 per gateway), reducing the need to deploy expensive edge-AI hardware at every location.



ENVIRONMENTAL BENEFITS



Supports early detection of forest fires at the initial ignition stage, reducing potential wildfire spread and ecological damage.



Helps preserve wildlife habitats by minimizing fire spread and human-wildlife conflicts.



Continuously monitors atmospheric pollutants (PM2.5, toxic gases) to support data-driven environmental protection policies.



1 EXISTING APPROACHES AND RESEARCH GAP

System / Project	Domain	Approach	Key Features	Limitations / Gaps
 Wildfire TinyML [1]	Forest fire	Arduino Nano 33 + BLE + environmental sensors; TinyML model	Real-time classification on MCU; extremely low-cost	Only lab demonstration (simulated data); limited outdoor validation; selected sensors only
 Industrial Fire (Uduremu et al.2025) [2]	Factory fire	ESP32-CAM + YOLOv5 + LoRaWAN/MQTT + (REM)	150 ms latency; zero packet loss under REM; multi-hop reliability	Tested in one lab/plant; focus on fires only; expensive camera hardware; not designed for forest/off-grid
 Air/Heat (NDMA SACHET App) [3]	Heatwave, Air quality	Mobile app + SMS / Cell Broadcast + CAP alerts	Real-time heat/dust alerts in Indian languages; leverages GPS/telecom	Dependent on government forecasts (IMD); does not measure local air quality; coverage only where app is used
 Multi-Hazard Prediction (2026) [4]	Multiple hazards	Sensor fusion + CNN-LSTM (edge-cloud)	94.3% overall accuracy; 3.5 sec alert latency (edge deployment)	Requires cloud infrastructure; not tested for long-term offline operation in remote areas
 IoT Environmental Monitoring (2026) [5]	Multi-environment	48-node network, MQTT/LoRaWAN, LSTM prediction	96.4% accuracy; 12 ms latency (MQTT); edge + cloud architecture	Demonstrated for specific regions; dependency on network; higher infrastructure complexity

4 KEY RESEARCH & REFERENCES

1

CWC Flood Forecast (2015) – Real-time flood forecasting using gauge and satellite data.

Central Water Commission (2015)

2

Developing a Fire Monitoring System Based on MQTT, ESP-NOW, and a REM in Industrial Environments.

Applied Sciences (2025)

3

Edge AI System for Real-Time and Explainable Forest Fire Detection.

SciTePress (2019)

4

An Edge Computing Environment for Early Wildfire Detection.

AeTIC (2022)

5

An Adaptive AI-Driven Multi-Hazard Prediction and Early Alert Framework Using Sensor Fusion and Deep Learning Models.

IEEE (2026)

6

IoT-Enabled Real-Time Environmental Monitoring and Disaster Prediction System.

IJAIML+ (2026)

2 PROJECT DIFFERENTIATION



Offline-First Operation
Works without continuous internet. Local sensing → LoRa → edge AI → GPS mapping → local dashboard.



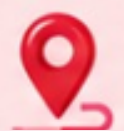
Sensor + Camera Fusion
Environmental anomalies trigger camera verification and AI classification for more reliable alerts.



Event-Triggered Edge AI
Lightweight sensing continuously; intensive visual inference only when risk is detected to save power.



Drift & Sensor Health Awareness
Monitors sensor baseline changes, calibration drift and abnormal readings to maintain reliability over time.



Geo-Tagged Threat Mapping
Every event is location-aware with GPS coordinates, sensor evidence and visual evidence for precise risk maps.



Shared Edge Gateway
One Jetson gateway can serve multiple sensor nodes ,reducing overall cost and enabling scalable deployment.

3 WHERE OUR SYSTEM FITS IN THE ECOSYSTEM



GOVERNMENT DATA & STANDARDS

7

Forest Survey of India (ISFR 2023) – Uttarakhand Forest Cover Data.

FSI (2023)

8

India Meteorological Department (IMD) – Weather and Forecast Data.

IMD

9

National Disaster Management Authority (NDMA) – SACHET Portal.

NDMA

10

C-DOT – Common Alerting Protocol (CAP) Integrated Alert System.

C-DOT / ITU-T X.1303



FULL REFERENCES

